

Questions:

1) Conditional branches: BEQ

- Stage 1 (IF/ID/RR): The instruction is fetched from instruction memory. It is decoded to identify it as a BEQ. $rs1$ and $rs2$ are read from Register file. The imm value is extracted and sign-extended.

- Stage 2 (EX/MEM/WB): ALU compares $rs1$ and $rs2$. If ALU zero flag is set, the branch target address is calculated ($PC + \text{offset}$). If they are not equal, $PC = PC + 4$.

2) Unconditional branch: JAL

- Stage 1: Instruction is fetched and decoded as JAL. The imm value representing jump offset is sign-extended.

- Stage 2: The return address ($PC + 4$) is written back to rd. $PC = PC + \text{immediate offset}$.

3) ALU: ADD

- Stage 1: Instruction is an R-type add instruction. The $rs1$ and $rs2$ are read from RF.

- ~~• Stage 2: The return address ($PC + 4$) is written back to rd.~~

- Stage 2: ALU performs addition on $rs1$ and $rs2$. The result is written into rd.

4) Load: LW

- Stage 1: Instruction is fetched and decoded as LW. The base address register $rs1$ is read from RF and immediate offset is sign-extended.

- Stage 2 ALU adds $rs1$ value and sign-extended imm to calculate mem address. The DMEM is accessed and the 32-bit used is written back to rd.

5) Store & SW's

- Stage 1 Both base address $rs1$ and register containing the data to be stored $rs2$ are read from RF. The imm offset is sign-extended.
- Stage 2 ALU calculates effective mem address by adding $rs1$ and imm offset. Value from $rs2$ is written to DMEM at calculated address. [No write back].